

Final PRACTICE Exam

MA 341

06/24/2021

Instructor Medvinsky Michael

Name: _____

Read all of the following information before starting the exam:

- Complete each problem. Show all of your work clearly and in order and justify your answers, as partial credit will be given when appropriate and there may be NO credit given for problems without supporting work.
- **Circle or otherwise indicate your final answers.** All answers should be completely simplified, unless otherwise stated.
- **You may use your own formula sheet (one sheet - two pages long) and a *non graphical* calculator.**

1.

Consider the following initial value problem.

$$\begin{cases} -\sin(x-y)dx + (3y^{-1}\cos(x-y) + \sin(x-y))dy &= 0 \\ y(0) &= \pi \end{cases}$$

- a. Is the equation is exact? Explain!
- b. If you found this equation not exact, find an equivalent exact equation.
- c. Solve the initial problem.

2. Find general solution of the following problems:

- a. $y'' + 16y = e^{-4x} + 3\sin 4x$
- b. $xy' + y = y^2x^2 \ln x$
- c. $y' + y \cot t = \frac{1}{\sin t}$

3. Solve the following initial value problem:

$$\begin{cases} y'' + 4y = f(t) = \begin{cases} \sin 3t, & 0 < t < \frac{\pi}{2} \\ 0, & t \geq \frac{\pi}{2} \end{cases} \\ y(0) = a \\ y'(0) = b \end{cases}.$$

4. Solve the following system of equations:

$$\begin{cases} y' - 4y - v &= t \\ v' - 2y - 3v &= -t \end{cases}$$

Good luck!